38 A scientist was asked to separate the following equations into two categories: nuclear fission and nuclear fusion.

1 $^{239}_{94}Pu \rightarrow {}^{112}_{46}Pd + {}^{124}_{48}Cd + 3{}^{1}_{0}n$

2 $^{3}_{2}He + {}^{3}_{2}He \rightarrow {}^{4}_{2}He + {}^{1}_{1}H + {}^{1}_{1}H$

3 $^{14}_{7}N + {}^{1}_{1}H \rightarrow {}^{15}_{8}O$

4 $^{235}_{92}U \rightarrow {}^{90}_{36}Kr + {}^{143}_{56}Ba + 2{}^{1}_{0}n$

Which equations show nuclear fission?

A 1 and 2 **B** 1 and 3 **C** 1 and 4 **D** 2 and 4